Government of Karnataka
Department of Technical Education
Board of Technical Examinations

Course: Thinking Programming with Python **Course Code: 25CS21I**

Time: 3 Hrs **Max Marks: 100**

Instructions: For Part-A questions, only the first written answers will be considered for evaluation.

PART A

I. Select the correct answer from the choices given 15x1 = 15

1. Which of the following is NOT a component of computational thinking?
   A) Abstraction B) Pattern Recognition C) Iteration D) Debugging
2. Which of the following best defines an algorithm?
   A) A programming language B) A step-by-step solution to a problem
   B) A type of loop D) A mathematical symbol
3. Which of the following is a valid variable name in Python?
   A) 2value B) value@ C) _value D) class
4. Which function is used to take input from the user in Python?
   A) read() B) get() C) input() D) scan()
5. What is the result of bool(0)?
   A) True B) False C) 0 D) Error
6. Which data type would you use to store "Hello, World!"?
   A) int B) float C) char D) string
7. You are asked to plan a method to find the top 3 scores in a class. What kind of problem is this?
   A) Searching B) Sorting C) Optimization D) Cryptographic
8. What is the role of abstraction in algorithm design?
   A) Ignoring unnecessary details and focusing on relevant data
   B) Executing the program faster
   C) Improving syntax
   D) Making the program run in reverse
9. What will be the output of: print("Hello" + "World")?
   A) Hello + World B) HelloWorld C) Hello World D) Error
10. Which operator has the highest precedence in Python?
    A) Addition (+) B) Multiplication (*)
    B) C) Exponentiation (**) D) Modulus (%)
11. Which keyword is used to skip the current iteration in a loop?
    A) skip B) continue C) break D) next

12. You are building an ATM system where the user must enter a 4-digit PIN and is allowed a maximum of 3 attempts. Which control structure is most suitable?
    A) for loop
    B) while loop with a counter
    C) if statement only
    D) match-case
13. How do you call a function named say_hello?
    A) call say_hello() B) say_hello() C) def say_hello() D) return say_hello()
14. What happens when break is used in the inner loop?
    A) Only the inner loop exits
    B) Both loops exit
    C) The program crashes
    D) The outer loop restarts
15. In an online shopping system, if the user's cart total exceeds ₹5000, a 20% discount is applied. Otherwise, a 10% discount is given. Which is the best Python construct to implement this?
    A) match-case
    B) nested loops
    C) if-else
    D) try-except

II. Fill in the blanks by choosing appropriate answer from those given in the bracket: (syntax, recursive, import, Searching,55,) 5X1 = 05

1. _____ problems involve finding a specific item in a dataset (e.g., searching for a name in a directory.
2. The set of rules that define how code is written in a language is called _____.
3. _____ is the output of the following? print("5" + str(5))
4. A function that calls itself is called a _____ function.
5. To rename the numpy module as np, you write: _____ numpy as np.

PART B

III. Answer any FIVE questions: 5X2 = 10

1. What is decomposition in the context of computational thinking?
2. Define abstraction in the context of problem-solving.
3. What are tokens in Python programming?
4. Name two loop control statements in Python.
5. Which function in Python helps you display output for debugging?
6. How would you define a syntax error in Python?

7. List two advantages of using functions in Python.
8. Name the two types of functions in Python.

PART C

VI. Answer any FIVE questions: **5X3 = 15**

1. Explain algorithmic thinking with a tooth brushing example.
2. Differentiate between abstraction and decomposition with suitable examples.
3. Explain the purpose of using variables in a Python program.
4. Discuss the importance of comments in Python and explain the difference between single-line and multi-line comments.
5. Describe how conditional statements work in Python. Give examples of if and if-else.
6. Explain with example how Python reports a syntax error.
7. How can a programmer systematically detect and fix semantic errors?
8. What is a nested loop in Python? Give one real-life example where nested loops are useful

PART D (Section I)

IV. Answer any FIVE questions: **5X5 = 25**

1. List and explain different types of problems that require algorithmic thinking. Provide one example for each.
2. Explain any two programming paradigms and compare their key features with suitable examples.
3. Explain the concept of type casting in Python. What is the difference between implicit and explicit type conversion? Provide examples.
4. Write python equivalent expressions for following
   i. f = a3 + b3 + 3ab(a + b)
   ii. Check if a number is positive
   iii. $S = a^3 + b^3 + 3ab(a + b)$
   iv. Calculate the average of three numbers a, b, and c
   v. Total Cost = Quantity × Unit Price – Discount
5. Describe the concept of debugging. What tools or strategies can a programmer use to identify and fix logical or semantic errors in a Python program?
6. The following Python code is meant to print all even numbers from 1 to 10. However, it contains some errors.

```
for i in range(1, 10)
  if i % 2 = 0:
    print i
  else
    continue
```

a. Identify and explain all syntax and logical errors in the code.

b. Rewrite the corrected version.

7. Explain the purpose of the return statement in Python functions. How is it different from print()? Write an example to illustrate.
8. Explain different ways to import and use built-in or user-defined modules in Python. Give examples for each.

PART D (Section II)

V. Answer any THREE questions: 10X3 = 30

1. Identify the tokens in the following Python statement and explain each of them with its category

```
if total_price >= 500.0:
    discount = total_price * 0.1  # Apply 10% discount
```

2. Create a program to manage airline ticket bookings based on the passenger's selected class: *"economy"*, *"business"*, or *"first"*.
   a. Prompt the user to input details such as name, age, destination, and class preference (*economy, business, or first*).
   b. Validate inputs (e.g., ensure the name contains only letters, age is a positive integer, and class selection is valid).
   c. Display appropriate error messages for invalid entries and allow the user to re- enter the data.
3. Create a Python program for a school that wants to automate the grading process based on a student's marks in three subjects:
   - Mathematics
   - Science
   - English

   Grading Policy:

Average Marks	Grade
≥ 90	A
75 ≤ average < 90	B
50 ≤ average < 75	C
< 50	F

Follow best programming practices (variable naming, indentation, and comments).

4. The following program is written to convert a temperature from Celsius to Fahrenheit and Kelvin and display the result. The code is written in a single block:

```
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9/5) + 32
kelvin = celsius + 273.15
print(f"Temperature in Fahrenheit: {fahrenheit:.2f} °F")
print(f"Temperature in Kelvin: {kelvin:.2f} K")
```

Rewrite this program using a **modular approach** by defining and using appropriate **functions** for:

a. Accepting input
b. Converting Celsius to Fahrenheit
c. Converting Celsius to Kelvin
d. Displaying the results
e. Organizing program flow using a main() function

5. Explain the concept of variable scope in Python. How are local and global variables handled inside functions? What is the use of the **global** keyword? Provide examples.

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